

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Information and Ideas	Command of Evidence	Hard

Question

Water flowing around an obstruction creates vortices (patterns of swirls) of varying size; by detecting the vortices, fish can determine the size and position of the obstruction. Testing by Yuzo R. Yanagisuru, Otar Akanyeti, and James C. Liao using models of three head shapes—narrow (low ratio of width to length), intermediate, and wide (high ratio of width to length)—showed that for medium-sized vortices, fish with wide heads would be least able to distinguish between vortices and general turbulence in the water. A second research team has therefore hypothesized that in low-visibility conditions, wider-headed fish will be less likely than narrower-headed fish to detect obstructions.

Which finding, if true, would most directly support the second research team’s hypothesis?

Answer

- A. A study using obstructions that created medium-sized vortices in low-visibility conditions found that the bristlemouth (*Chaetostoma yurubiense*), which has a relatively wide head, bumped into more than half of the obstructions.
- B. A study using obstructions that created medium-sized vortices in low-visibility conditions found that some specimens of dusky smooth-hound (*Mustelus canis*), which has a relatively narrow head, bumped into the obstructions more often than other specimens of the same fish did.
- C. A study using obstructions that created medium-sized vortices in low-visibility conditions found that the wider-headed bristlemouth (*Chaetostoma yurubiense*) bumped into obstructions more often than the narrower-headed dusky smooth-hound (*Mustelus canis*) did.
- D. A study using obstructions that created medium-sized vortices in low-visibility conditions found that the narrower-headed dusky smooth-hound (*Mustelus canis*) bumped into the obstructions just as often as the wider-headed bristlemouth (*Chaetostoma yurubiense*) did.

Correct Answer: C

Rationale

Choice C is the best answer because it presents a finding that, if true, would most directly support the second research team’s hypothesis about how a fish’s head shape affects the fish’s ability to detect obstructions. The text explains that fish detect obstructions using water vortices and that testing by Yanagisuru, Akanyeti, and Liao showed that fish with wide heads struggle to distinguish medium-sized vortices from general turbulence in the water. Based on this finding, a second research team hypothesized that in low-visibility conditions, wider-headed fish would be less effective at detecting obstructions than narrower-headed fish. The finding that wider-headed bristlemouth fish bumped into obstructions more often than narrower-headed dusky smooth-hound fish did under the specified conditions (low visibility with medium-sized vortices) would directly support the second team’s hypothesis.

Choice A is incorrect because it would provide only partial support for the second research team’s hypothesis. While this finding shows that wide-headed bristlemouth fish bumped into many obstructions, it provides no comparison with narrow-headed fish. The hypothesis specifically predicts a difference in detection ability between fish with wider heads and those with narrower heads, so supporting evidence would need to include data from both head types to allow for this comparison. Choice B is incorrect because it would not support the second research team’s hypothesis comparing obstruction detection between fish with wide heads and those with narrow heads. Instead, it examines variation within a single narrow-headed species, which would not address the hypothesis about differences between fish with wider and narrower head shapes. Choice D is incorrect because this finding would contradict rather than support the second research team’s hypothesis. The hypothesis predicts that wider-headed fish will be less likely to detect obstructions than narrower-headed fish in low-visibility conditions, which would result in wider-headed fish bumping into obstructions more frequently. A finding showing equal bumping rates between fish with the two head shapes would indicate no relationship between head width and detection ability, directly contradicting the hypothesis.